PathwayAI | Dutchess County Lyme Burden & Action Report

PathwayAI: Making the Hidden Patient Journey Visible—and Actionable

Public evidence and support planning · Version 1.2

Prepared: 2026-11-01

Data years: Lyme surveillance 2024 · HRSA insurance 2022 · HRSA poverty and resources 2023

Original study-based cost illustration: USD 2,153,920 in 2016 dollars.

Optional 2025 purchasing-power comparison: USD 2,889,245 · 11 available months; general inflation.

Illustrations of scale, not measured county spending.

1. OBSERVED PUBLIC HEALTH DATA

Reported Lyme cases: 1,060 · 2024 · county FIPS 36027.

Reported rate: 355.5 per 100,000 residents. New York State: 109.8.

NYSDOH reports cases by residence, not all infections or where exposure occurred. CDC historical county files in the app cover 2019–2022; they are not combined with 2024 counts.

Published rates are reproduced; exact population denominators and their vintage have not been independently verified.

Healthcare resource context · HRSA AHRF 2024–2025 release · 2023 fields

Primary-care physicians: 233

Hospitals: 2

Hospital beds: 404

Existing public AHRF asset; 2023 county fields.

General healthcare resource context, not Lyme expertise, appointment availability, interpreter access or a verified shortage.

COUNTY COMPARISONS AND PLANNING CONTEXT

Reported surveillance · 2024

Dutchess: 1,060 reported cases; 355.5 per 100,000 residents.

Orange: 964 reported cases; 237 per 100,000 residents.

New York State: 21,632 reported cases; 109.8 per 100,000 residents.

Dutchess reported-rate trend · cases per 100,000

2022: 394.6

2023: 358.9

2024: 355.5

Insurance and poverty context · HRSA AHRF 2024–2025

Uninsured residents under age 65: 4.7% · 2022 · Residents under age 65; AHRF field population

People below poverty level: 8% · 2023 · Population for whom poverty status is determined

Different field years; not Lyme-attributable burden or measured demand. Margins of error are not supplied in these fields.

Reported disease by residence, not exposure location or all infections. Trend years use different source publications; do not infer causes, changes in access or savings. Pre-2022 rates are excluded because surveillance definitions changed. Published 2024 rates are reproduced without reconstructing population denominators.

Language other than English spoken at home: 16.2%

Residents age 5 and older · ACS 2020–2024

Existing app QuickFacts snapshot; not freshly retrieved.

Not a measure of English proficiency, immigration status or interpreter demand.

Existing county initiatives to check

Tick Task Force · Dutchess County · Tick-borne disease prevention and education

https://www.dutchessny.gov/Departments/County-Legislature/Tick-Task-Force.htm

Mobile Health Unit · Dutchess County Department of Behavioral & Community Health · Community outreach and access

https://www.dutchessny.gov/Departments/DBCH/Mobile-Health-Unit.htm

Community Health Improvement Plan · Dutchess County Department of Behavioral & Community Health · Community health planning

https://www.dutchessny.gov/Departments/DBCH/Community-Health-Improvement-Plan.htm

Existing public directory snapshot; current contacts, activity, eligibility and availability must be verified. No listing does not mean no services exist.

Questions these indicators can help you ask

- How do the reported rate and case count compare with the neighboring county and state?

- Which prevention and outreach questions should residents and partners explore in light of the reported-rate trend?

- Which languages and communication formats do residents actually prefer?

- Could existing county initiatives help, and what staffing, service capacity or support gaps need checking?

2. STUDY-BASED COUNTY COST ILLUSTRATION

2025 purchasing-power illustration: 1,060 reported cases × inflation-adjusted study mean ≈ USD 2,889,245.

Inflation-adjusted study mean: approximately USD 2,726 per participant (rounded for display).

General inflation comparison using 11 available 2025 months; calculations use unrounded values. Original source calculation and study population are in Methods.

Societal: medical, nonmedical and productivity costs; components are already included.

Applying this study mean to reported county cases is an assumption; the result is not measured local spending.

No adjustment for underreporting or local case mix. No causal savings or disability probabilities are inferred.

This illustrates scale, not today's local costs. Hook excluded non-English speakers and nonacute disease; the study does not represent every local household.

Optional 2025 purchasing-power comparison · 11 available months

2025 purchasing power: USD 2,726 per participant; county illustration: USD 2,889,245.

General inflation only. 2016 mean CPI: 240.007; 2025 available-month mean: 321.943; factor approximately 1.34.

Calculations use the full-precision factor retained in the methods CSV.

2025 has 11 observed months; October is missing. No imputation. This is not a verified official annual-average restatement.

Inflation changes purchasing power, not treatment patterns or local representativeness.

https://fred.stlouisfed.org/series/CPIAUCNS

Source workbook SHA-256: 346ef77b0f6cc7bb59c42d27be2c802a5aefb3f518339191e9320a1cb105d28a

3. PATIENT SURVEY FINDINGS · NOT COUNTY ESTIMATES

Snapshot: 2026-10-03 · 33 respondents

Small, self-selected sample across locations; respondents self-reported a Lyme diagnosis.

The supplied summary raises questions about spending, time lost, repeated visits and diagnosis delays.

Not representative of the selected county or the U.S. population. Raw responses, question-specific denominators and cost/time reporting periods have not been verified here; no survey percentages or county totals are calculated.

Project-lead-supplied summary already documented in the app; not a new analysis of raw responses.

Only this aggregate context is included. Personal app entries are not included in community evidence.

What we still need to understand

- Lyme-attributable county disability and duration
- Continuing costs beyond the study follow-up
- Local unmet care and support needs
- Experiences of people excluded or underrepresented in the source study

4. FROM LYME DISEASE BURDEN TO COMMUNITY ACTION

Suggested actions for local review, not clinical advice or adopted programs.

Prevention education and tick-bite awareness: Share established prevention information through trusted local channels.

Timely access to healthcare: Help residents find care and ask about appointments, costs, transport and interpreters.

Financial and caregiving burden: Recognize spending, missed work, unpaid care and support needs beyond medical bills.

Language access and community resources: Make information easier to understand and connect residents with existing local support.

Planning audience: Public health departments

- Share tick-prevention information through local events and trusted community channels.

- Review county surveillance alongside service availability; ask residents which barriers need attention.

- Bring residents and potential partners together to choose one support priority and agree how to follow up.

5. A SUPPORT QUESTION TO DISCUSS

Getting to care

Planning question: Should a county care-navigation team investigate a small ride-navigation pilot with willing residents?

Possible response: Ask willing residents and transport providers to review ride eligibility, accessibility and appointment coordination.

Who could help: Patient and caregiver representatives, transport services and care navigators

These are proposed discussion topics, not findings about local demand or commitments from partners.

6. WHAT TO CHECK BEFORE STARTING A PILOT

- Confirm local demand with residents and caregivers, including assisted/offline participation.

- Verify current services, eligibility, interpreter access and appointment capacity.

- Agree who is responsible and estimate staff time, program costs and a funding source.

- Define consent, minimal data collection, access, retention and deletion before any collection.

7. CHECK WHETHER PEOPLE RECEIVED SUPPORT

Choose a common follow-up period, such as 30 days. Record the starting situation and support received. Keep unanswered questions marked as unknown.

For this priority, check: Requested and completed accessible rides; appointments attended; travel spending; support still needed

Use findings to improve support. Before/after changes alone do not establish that the tool caused improvements or saved money.

8. DATA SOURCES AND METHODS

Source status: Existing source snapshots. Newer official releases have not been verified for this brief.

Original study basis (retained for traceability)

1,060 reported cases × USD 2,032 study mean = USD 2,153,920 (2016 dollars).

Hook et al. (published 2022): 613 participants in societal analysis; reported cases recruited in four high-incidence states during 2014–2016.

Reported cases, county of residence · 2024

https://www.health.ny.gov/statistics/diseases/communicable/2024/docs/cases.pdf

Previously checked supplied official PDF, page 5.

Reported rates per 100,000 · 2024

https://www.health.ny.gov/statistics/diseases/communicable/2024/docs/rates.pdf

Previously checked supplied official PDF, page 5.

CDC surveillance interpretation; historical county file in app · 2019–2022 county file; separate from NYSDOH 2024

https://www.cdc.gov/lyme/data-research/facts-stats/index.html

Background surveillance context, not an additional 2024 county estimate.

Healthcare resources, insurance and poverty context · 2022 insurance fields; 2023 healthcare and poverty fields; AHRF 2024–2025 release

https://data.hrsa.gov/topics/health-workforce/nchwa/ahrf

Existing public AHRF asset; 2023 county fields.

Societal study mean · 2016 dollars; recruitment 2014–2016

https://wwwnc.cdc.gov/eid/article/28/6/21-1335_article

Checked user-supplied public article text: Methods, Results and Discussion. Linked tables not separately retrieved. Live website blocked.

Reported rate trend, Figure 147, page 219 · 2022–2023; regional assessment published 2025

https://www.dutchessny.gov/Departments/DBCH/Docs/MHRCHA2025.pdf

Existing source snapshot; no fresh retrieval.

Language spoken at home, age 5+ · ACS 2020–2024

https://www.census.gov/quickfacts/dutchesscountynewyork

Existing source snapshot; no fresh retrieval.

Public evidence and proposed planning actions only. No session stories, personal records or synthetic patient cards are included.

Discussion brief, not a county demand assessment, benefits decision, funded service or adopted county plan.